YOUTH UNEMPLOYMENT DATA VISUALIZATION DASHBOARD WITH STORYTELLING

1. Project Overview

This project builds an interactive, responsive dashboard in Streamlit for India's youth (age 15-29) unemployment data from the Periodic Labour Force Survey (PLFS). Instead of presenting charts without context, the dashboard follows a story: understand the headline rate, see whether it is improving, find which education groups are hit hardest, check who within those groups is most affected, and decide which states need attention first.

Objectives

- Summarise youth unemployment with headline KPIs (latest rate, change, urban-rural gap, graduate rate).
- Track the national trend from 2020-21 to 2025 and flag where the survey reference period changes.
- Compare unemployment across education levels and expose the education paradox.
- Compare male and female rates by education and by rural/urban sector.
- Compare 29 states against the national rate.
- Allow users to filter the data and download the filtered records.

Technology Stack

Python, Streamlit, Pandas and Plotly. The dataset is bundled with the app; there is deliberately no file-upload option.

2. Dataset Description

The supplied CSV (**youth_unemployment_india_plfs.csv**) contains 136 records of one metric: Youth (15-29) unemployment rate, usual status (ps+ss), expressed as a percentage. Records cover six periods (2020-21 to 2023-24 on a July-June basis, 2024 and 2025 on a January-December basis), 29 states plus All India, ten education levels, sex (Male, Female, Everyone) and sector (Rural, Urban, Rural+Urban).

Expected columns: Period, Period_Basis, Geography, Education_Level, Sex, Sector, Metric, Value_Percent, Source.

Slice	Records	What it supports
National headline, 2020-21 to 2025	6	Trend charts and KPIs
All India, 2025: education x sex x sector	99	Education, gender and sector views, heat-map
States, 2023-24 (all levels)	29	State comparison
2025 rural / urban headline	2	Urban-rural gap KPI

Important: the data are real published PLFS figures, but they are not one uniform panel. Education and rural/urban detail exist for 2025 only, state detail exists for 2023-24 only, and the reference period changes between 2023-24 and 2024. The dashboard labels these limits on the relevant pages.

3. Dashboard Design and Storytelling Flow

The app has six views, reached from a navy sidebar that also holds the filters. Every view repeats the same four KPI cards, then one or two charts, then a **Story** statement (a headline, a data-driven observation and a suggested action), followed by a **Decision Prompt** or **Data Caution** note.

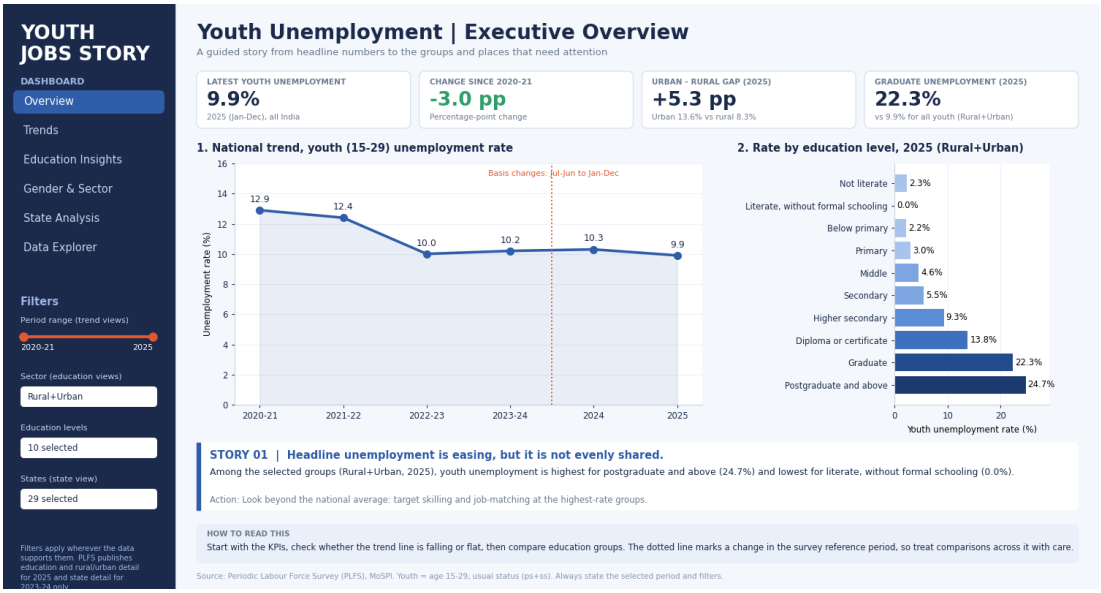
View	Question it answers	Visuals
Overview	What is the headline, and who is affected?	KPI cards, area/line trend, ranked horizontal bars
Trends	Is it getting better?	Line chart with basis-change marker, change bars, rural vs urban bars
Education Insights	Does more schooling mean more jobs?	Ranked bars, male/female/overall ladder lines
Gender & Sector	Who is hardest hit?	Grouped bars, diverging gap bars, heat-map
State Analysis	Where is the pressure?	Sorted bars with national line, donut, top/bottom-5 cards
Data Explorer	What is the evidence?	Filterable table, KPIs, CSV download

Responsive design: KPI cards use a CSS auto-fit grid that wraps from four across to one column on a phone; chart columns stack automatically on narrow screens; Plotly charts stretch to container width with legends above the plot; the sidebar collapses on mobile.

4. Dashboard Screenshots

The following screenshots are illustrative dashboard mockups. They were rendered with Matplotlib from the same data-preparation code (utils.py) that powers the Streamlit app, so every number, ranking and story sentence matches the app. The live Streamlit application is interactive, and its Plotly styling, fonts and spacing will differ slightly depending on screen size and the filters selected.

Screenshot 1 — Executive Overview



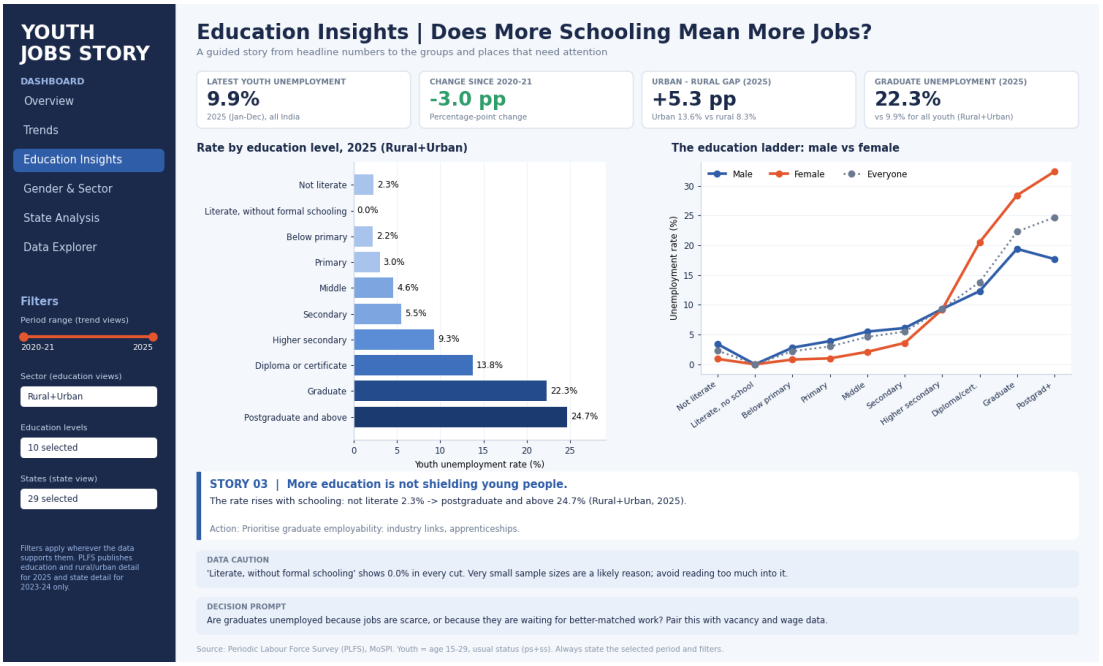
The opening view combines four KPIs with the national trend and a ranked comparison of education levels. The story statement translates the charts into a business implication: the national rate fell from 12.9% to 9.9%, but graduates are at 22.3%.

Screenshot 2 — Trends



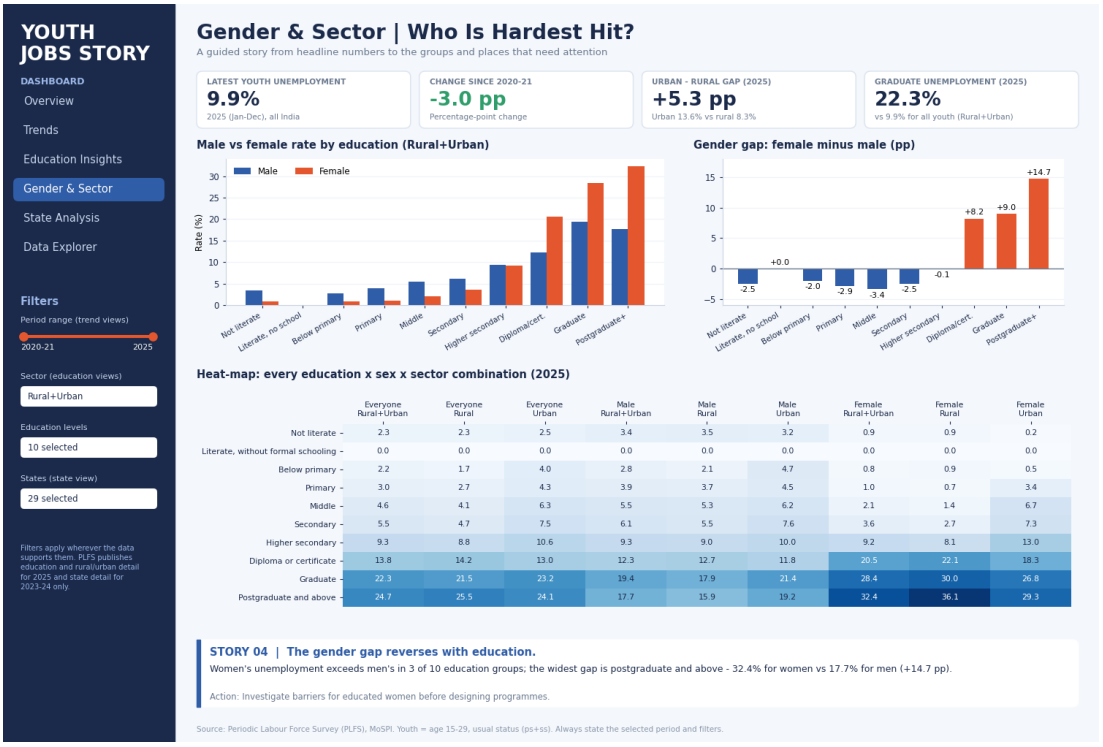
The trend view shows the rate over time with a dotted marker where the survey changes from July-June to January-December years, plus period-on-period change and the 2025 rural/urban split. Most of the improvement happened by 2022-23; since then movement is within a few tenths of a point.

Screenshot 3 — Education Insights



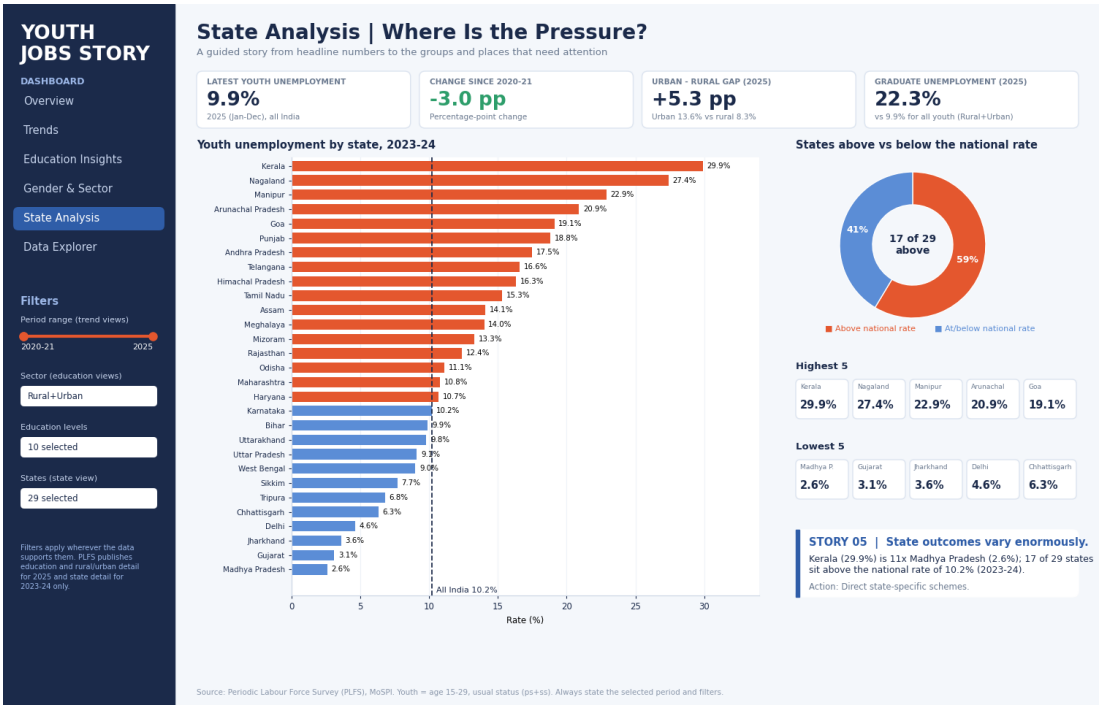
Unemployment rises almost steadily with schooling, from about 2% for the least educated to 24.7% for postgraduates. The ladder chart adds a male/female split, showing the lines crossing at higher secondary level.

Screenshot 4 — Gender & Sector



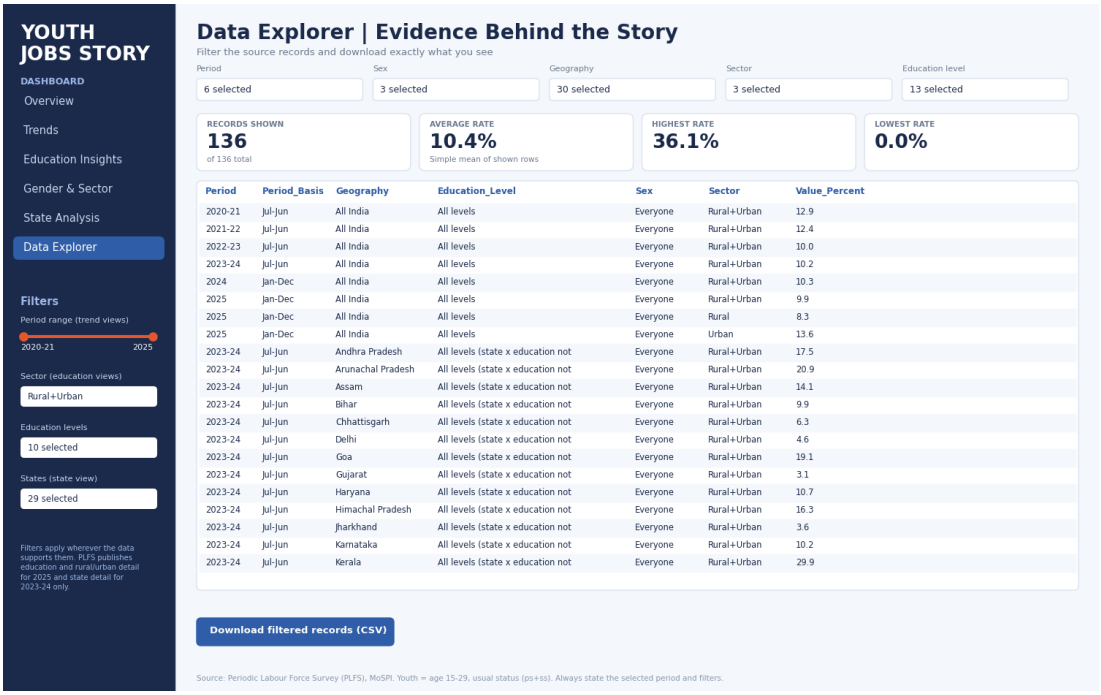
Women's rates are lower than men's up to secondary schooling but far higher from diploma level upward (postgraduate: 32.4% vs 17.7%). The heat-map lets users scan every education x sex x sector combination; rural women with postgraduate degrees show the highest value in the dataset (36.1%).

Screenshot 5 — State Analysis



State bars are coloured by whether they sit above the national 2023-24 rate (10.2%). Seventeen of 29 states do; Kerala (29.9%) and Nagaland (27.4%) lead, while Madhya Pradesh (2.6%) and Gujarat (3.1%) are lowest.

Screenshot 6 — Data Explorer



The explorer exposes the underlying records with filters for period, sex, geography, sector and education level, and a button that downloads exactly the filtered rows as CSV.

5. Visualisation Methods Used

- **KPI cards** (as in the reference dashboard) for the headline numbers; the change KPI turns green when the rate falls.
- **Line / area chart** for the national trend, with a dotted line marking the change in survey reference period.
- **Horizontal bar chart** with a blue gradient (darker = higher) for ranking education levels and states.
- **Column chart** for rural vs urban and for period-on-period change (green = improvement, orange = rise).
- **Donut chart** (the reference used a pie) for the share of states above the national rate.
- **Added storytelling methods:** grouped bars and a diverging gap chart for gender; a heat-map for education x sex x sector; a reference line at the national rate on the state chart; top-5 / bottom-5 cards; and a dynamic Story statement whose numbers are calculated from the data and filters.

6. How to Run

Project files: app.py (UI and charts), utils.py (data logic), data/youth_unemployment_india_plfs.csv, requirements.txt and .streamlit/config.toml (light theme).

```
pip install -r requirements.txt
streamlit run app.py
```

The app opens at http://localhost:8501. Use the sidebar to switch views and apply filters.

7. Filters and Their Scope

- **Period range:** applies to the national trend, KPIs and change charts.
- **Sector:** applies to the education, gender and overview views (2025 data).
- **Education levels:** applies to the education-based charts, story text and heat-map.
- **States:** applies to the state view. State data exist only for 2023-24 and have no education breakdown.
- **Data Explorer:** has its own five filters and a CSV download of the filtered records.

8. Storytelling Guidelines

- Lead with a question, not a chart title alone.
- Describe the observed pattern, then explain why it may matter.
- Separate observations from hypotheses; verify causes with labour-market context.
- Show male and female results side by side; an overall average hides the largest gaps.
- Always state the selected period, sector and filters when presenting results.
- Do not compare across the July-June / January-December boundary without noting it.
- Treat zero or very small values (e.g. 'Literate, without formal schooling' = 0.0%) as possible small-sample effects.

9. Conclusion

The dashboard turns PLFS youth unemployment records into a sequence of questions and potential actions. Its filters let users move from the national headline to education, gender, sector and state evidence. The central message is that the national rate (9.9% in 2025) hides sharp differences: unemployment is highest among graduates and especially educated women, and varies roughly tenfold between states. These are descriptive patterns; explaining their causes needs additional data on job vacancies, wages and labour-force participation.